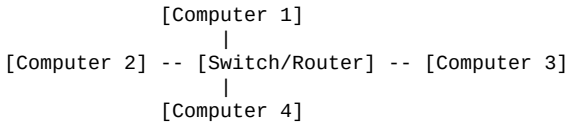
TASK: 1

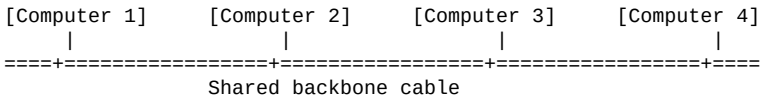
Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

A. Star Topology



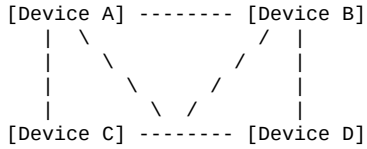
Example: A Flipkart warehouse can use star topology because each computer connects to a central switch, making management and troubleshooting easier.

B. Bus Topology



Example: A small older local network could use bus topology because devices share one backbone cable, though a cable fault may interrupt the whole network.

C. Mesh Topology



Example: A mesh network can connect important servers or wireless mesh access points because multiple paths may keep communication working if one link fails.

NETWORK

TASK: 2

Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Topology	Pros (4)	Cons (4)	Best Use Case / App Example
Star	1. Easy to install/manage. 2. Easy to locate faults. 3. One device failure usually does not affect others. 4. Easy to add devices.	1. Switch failure can stop network. 2. Uses more cable than bus. 3. Switch adds cost. 4. Limited by switch ports.	Best: offices, labs, warehouses. Example: Flipkart-style order processing.
Bus	1. Uses less cable. 2. Low cost for a tiny setup. 3. Simple layout. 4. No central switch needed.	1. Backbone failure can stop network. 2. Performance drops with traffic. 3. Faults are hard to locate. 4. Limited scalability/collisions.	Best: small or legacy networks. Example: basic shared data network; not ideal for modern Instagram-scale services.
Ring	1. Predictable data path. 2. Devices forward signals. 3. Token rings can avoid collisions. 4. Orderly access to network.	1. A break can disrupt a basic ring. 2. Changes may interrupt service. 3. Troubleshooting can be difficult. 4. Less flexible than star for small LANs.	Best: specialized/legacy ring systems. Example: some industrial/provider networks carrying data for services such as Zomato.
Mesh	1. Multiple paths between nodes. 2. High reliability. 3. Can reroute around failed links. 4. Good for critical connections.	1. Expensive to install. 2. Needs many links/ports. 3. Complex to configure. 4. Harder to maintain at scale.	Best: resilient backbones/wireless mesh. Example: mesh links can help maintain connectivity for services such as Paytm.

Note: Apps such as Instagram, Zomato, and Paytm use distributed servers and networks; an app does not depend on only one physical topology. These are examples of where each topology could be useful.

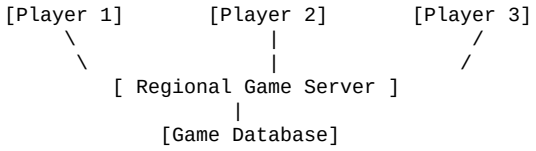
NETWORK

TASK: 3

Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

Chosen topology: Star / client-server topology

I would use a client-server star topology, where every player's phone connects to a central game server or a nearby regional server. This lets the server coordinate player positions, actions, and game updates quickly, reducing delays when the server is close to players. Reliable servers and backup connections improve availability if a link fails. It costs more than a simple peer-to-peer setup, but is easier to control, secure, and maintain for a competitive multiplayer game.



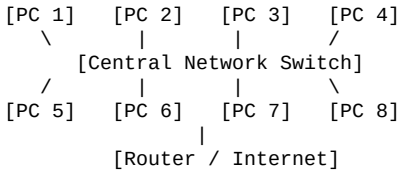
Impact: Speed: low latency with nearby servers. Reliability: backup links and reliable servers help keep matches stable. Cost: moderate to high because servers and maintenance are needed.

NETWORK

TASK: 4

Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines. Hint: Consider factors like cost, ease of troubleshooting, and future expansion.

Recommended topology: Star topology



Star topology is suitable because all eight computers connect to one central switch, which is affordable and easy to manage. If one computer or its cable fails, the others can continue working, and more computers can be added to available switch ports.

Conclusion: Star topology is the best practical choice for the warehouse because it supports reliable order processing, easy troubleshooting, and future expansion.

NETWORK